

Dual Decomposition

$$\text{Primal: } \min: f(x) + g(Ax)$$

$$\text{Dual: } \max: -f^*(-A^T z) - g^*(z)$$

When could we gain a computational advantage by working in the dual.

- Dual may involve more simple calculations - example: no constraints hence exploit some additional structure.
- Exploit smoothness in the dual
- separable problem in dual but not primal.

Algorithms in the Dual

- Gradient descent (ascent)
- Proximal Point Alg.
- Proximal Gradient
- Variants

Dual Gradient Ascent - Equality Constraints

$$\begin{array}{ll} \min: f(x) & \longleftrightarrow \max_z: \underbrace{-f^*(-A^T z) - b^T z} \\ \text{st: } Ax = b \end{array}$$

$$\text{Gr. Ascent: } z_{t+1} = z_t + \lambda \underbrace{\nabla (-f^*(-A^T z) - b^T z)}$$

$$\text{subgrad} \in \partial (-f^*(-A^T z) - b^T z) = -b + \underbrace{A \partial f^*(-A^T z)}_{\hat{x}}$$

$$\Rightarrow z_{t+1} = z_t + \lambda (A \hat{x} - b)$$

$$\hat{x} \in \partial f^*(-A^T z) \Leftrightarrow -A^T z \in \partial f(\hat{x})$$

$$\Leftrightarrow 0 \in \partial f(\hat{x}) + A^T z$$

$$\Leftrightarrow \hat{x} \in \underset{x}{\text{argmin}}: f(x) + z^T A x - z^T b$$

$$\text{GA: } z_{t+1} = z_t + \lambda (A \hat{x} - b)$$

$$\hat{x} \in \underset{x}{\text{argmin}}: f(x) + z^T A x$$

Dual Gradient Ascent - Separable Problem

$$(P) \quad \begin{array}{l} \min: f_1(x_1) + f_2(x_2) \\ \text{st: } A_1 x_1 + A_2 x_2 \preceq b \end{array} \quad // \text{ coupling or complicating}$$

$$(D) \quad \begin{array}{l} \max: -f_1^*(-A_1^T z) - f_2^*(-A_2^T z) - b^T z \\ \text{st: } z \succeq 0 \end{array} \quad / \text{ Check}$$

Dual Gradient Ascent - Separable Problem - continued -

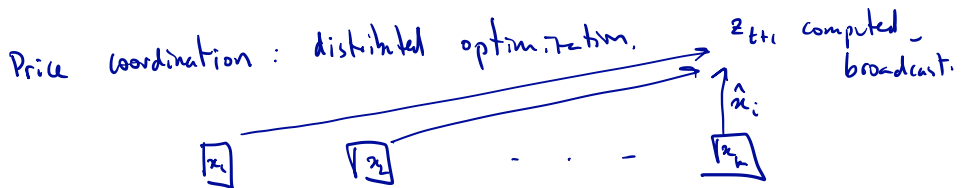
$$\max: -f_1^*(-A_1^T z) - f_2^*(-A_2^T z) - b^T z$$

$$\text{st: } z \in \mathbb{R}^n$$

$$z_{t+1} = z_t + \lambda \text{Proj}_{\mathbb{R}^n} \left\{ A_1 \hat{x}_1 + A_2 \hat{x}_2 - b \right\}$$

$$\hat{x}_1 \in \partial f_1^*(-A_1^T z_t), \quad \hat{x}_2 \in \partial f_2^*(-A_2^T z_t)$$

$$\Rightarrow \begin{aligned} \hat{x}_1 &\in \arg\min: f_1(x_1) + z^T A_1 x_1 \\ \hat{x}_2 &\in \arg\min: f_2(x_2) + z^T A_2 x_2 \end{aligned} \quad \Bigg/ \text{ decoupled$$



Separable Problem - Example -

$$\min: \sum_i \left(\frac{1}{2} x_i^T P_i x_i + q_i^T x_i \right)$$

$$\text{st: } B_i x_i \preceq d_i \quad \forall i$$

$$\sum A_i x_i \preceq b \quad \leftarrow \text{one coupling constraint.}$$

Exercise: Compute dual $\hat{c}_i$ compute what
Gradient Ascent looks like.

Compare to naive complexity of solving
primal directly.